Curriculum Vitae – Georgios Kissas

Personal Information

Georgios Kissas
OAT X11, Andreasstrasse 5, 8092 Zürich, Switzerland
+41 789699577
gkissas@ai.ethz.ch
www.linkedin.com/in/georgioskissas/
scholar.google.com/georgioskissas

Awards

- **Gold Reviewer** ICML, 2026
- **EUROMECH Colloquium 662** Early Career Invited Talk Award, 2026
- **AWS Free Credit** Trainium TPU Servers, 2025
- **Top Reviewer** NeurIPS, 2024
- **Asuera Stiftung Scholarship** ETH Zürich Foundation, 120000 CHF, 2023
- **Post Doctoral Fellowship** ETH AI Center (1.5% acceptance rate), 2023
- **Outstanding TA Award** MEAM UPenn, 2021
- **Gerondellis Foundation** Scholarship Award, 2020
- **MIT Travel Grant** Learning for Dynamical Systems and Control, 2019
- **Doctoral Fellowship Award** University of Pennsylvania, 2018
- **Limmat Stiftung Award** Award for Academic Excellence, 2017

Professional Experience

Senior Data Scientist, Research Swiss Data Science Center *November 2025-Present*

- Development of novel deep learning algorithms for scientific discovery using language modeling, large scale AI with applications to Exascale Cosmological Astrophysics simulations, and industrial CFD applications.

Post Doctoral Researcher & Lecturer ETH Zurich *June 2023-November 2025*

- Joint Post Doctoral positions between the Applied Mathematics, Engineering and Computer Science.
- Development of novel deep learning algorithms for scientific discovery through language modeling, and first-principles AI with applications to dynamical systems.

Research Assistant University of Pennsylvania, PI Lab *August 2018-May 2023*

- Worked in the areas of Deep Learning for physics simulations and precision medicine, Operator Learning, Fluid Dynamics for Cardiovascular Flows, Bayesian Inference, Deep Learning for Dynamical Systems and Sensitivity Analysis.

Technische Angestellte Wacker Chemie AG *June 2018-August 2018*

- Developed and enhanced analytical and computational models for modeling solid liquid two-phase flows and computing the dimensions of fractal nanoparticles and the effective viscosity of suspensions.

Research Assistant National Technical University of Athens, COMSE Group *May 2016-May 2018*

- Developed a Finite and Spectral Element solvers for the system of equations derived from Self-Consistent Field Theory.

Education

University of Pennsylvania, Philadelphia, PA, USA

PhD, Mechanical Engineering and Applied Mechanics, May 2023
Thesis: *"Towards digital twins for cardiovascular flows: A Hybrid Machine Learning and Computational Fluid Dynamics Approach"*
Advisor: Paris Perdikaris

National Technical University of Athens, Athens, Greece

M.S., Computational Mechanics (Fluids), May, 2017, (2nd Highest GPA)
Thesis: *"A computational study of Self-Consistent Field Theory for polymer interfaces"*
Advisor: Doros Theodorou

University of Patras, Patras, Greece

Diploma, Mechanical and Aeronautical Engineering, February, 2015 (Honors)
Thesis: *"A comparative study of Wavelet and Hilbert-Huang Transformations for fault detection in slew ring bearings"*

Patents

Computer Systems and Methods for Learning Operators, P. Perdikaris, G.J. Pappas, J. Seidman, **G. Kissas** - Provisional Patent Application 22-9935 (https://www.freepatentsonline.com/y2023/0214661.html).

Publications

Brivio, S, **Kissas, G.**, Manzoni, A., and Mishra, S., 2026, Reduced-Order Operator Learning through Perceivers: Dissecting the Hidden Mechanisms. (*To appear in the ArXiv*)

Koren, N., Hofmann, T., **Kissas, G.**, METRO: Metric-Enhanced Token Routing Operator. (Under Review NeurIPS 2026)

Koren, N., Hofmann, T., **Kissas, G.**, Advectra: Asymmetric Latent Transport for Non-Stationary Physics. (Under Review NeurIPS 2026)

Kissas, G., Bonev, B., Oikonomou, O., Lingsch, L., Chamon, F. O. L., 2026, Attention, Generators, and Semigroups: Structurally Stable Deep Transformers. (Under Review NeurIPS 2026)

Yu, K., Chatzi, E. and **Kissas, G.**, 2026. Neuro-Symbolic ODE Discovery with Latent Grammar Flow. arXiv preprint, arXiv:2604.16232.

Lingsch, L., **Kissas, G.**, Jakubik, J., Mishra, S., 2026, Phaedra: Learning High-Fidelity Discrete Tokenization for the Physical Science, arXiv preprint, arXiv:2602.03915

Oikonomou, O., Lingsch, L., Grund, D., Mishra, S. and **Kissas, G.**, 2025. Neuro-Symbolic AI for Analytical Solutions of Differential Equations. Forty-Third International Conference on Machine Learning, 2026

Renner, D. Mishra, S., and **Kissas, G.**, 2026. Accelerated Patient-Specific Calibration via Differentiable Hemodynamics Simulations. arXiv preprint arXiv:2412.14572.(*Under Review ACM TOMS*)

Yu, K., Chatzi, E. and **Kissas, G.**, 2025. Grammar-based Ordinary Differential Equation Discovery. Mechanical Systems and Signal Processing, 240, p.113395.

Sokolow, R., Bhattacharya, A., **Kissas, G.**, Thompson, E.W., Beeche, C., Swago, S., Zhang, D., Viswanadha, M., Morse, C., Chirinos, J. and Damrauer, S., 2025. Reduced order computational fluid dynamic simulations in the thoracic aorta are associated with disease recorded in a medical biobank. Scientific Reports, 15(1), p.39203.

Lingsch, L.E., Grund, D., Mishra, S. and **Kissas, G.**, 2024. FUSE: Fast Unified Simulation and Estimation for PDEs, *Thirty-eighth Conference on Neural Information Processing Systems*, 2024.

Kissas, G., Mishra, S., Chatzi, E. and De Lorenzis, L., 2024. The Language of Hyperelastic Materials. *Computer Methods in Applied Mechanics and Engineering*, 428, p.117053.

Seidman, J.H., **Kissas, G.**, Pappas, G.J. and Perdikaris, P., 2023. Variational Autoencoding Neural Operators, *International Conference on Machine Learning*, 2023.

Kissas, G.*, Seidman *, J.H., Guilhoto, L.F., Preciado, V.M., Pappas, G.J. and Perdikaris, P., 2022. Learning Operators with Coupled Attention. *Thirty-sixth Conference on Neural Information Processing Systems, Journal to Conference Track* (***Spotlight***)

Seidman, J.H.*, **Kissas, G.** *, Perdikaris, P. and Pappas, G.J., 2022. NOMAD: Nonlinear Manifold Decoders for Operator Learning, *Thirty-sixth Conference on Neural Information Processing Systems.* (***Featured in the "Paper of the Week" section of DeepAI***)

Kissas, G.*, Hwuang, E., Thompson, E.W., Schwartz, N., Detre, J.A., Witschey, W.R. and Perdikaris, P., 2022. Feasibility of Vascular Parameter Estimation for Assessing Hypertensive Pregnancy Disorders. Journal of Biomechanical Engineering.

Yang, Y., **Kissas, G.**, Perdikaris, P., 2022. Scalable uncertainty quantification for deep operator networks using randomized priors. *Computer Methods in Applied Mechanics and Engineering*, 399, p.115399.

Kissas, G.*, Seidman *, J.H., Guilhoto, L.F., Preciado, V.M., Pappas, G.J. and Perdikaris, P., 2022. Learning Operators with Coupled Attention. *Journal of Machine Learning Research*, 23(215), pp.1-63

Kissas, G., Yang, Y., Hwuang, E., Witschey, W.R., Detre, J.A. and Perdikaris, P., 2020. Machine learning in cardiovascular flows modeling: Predicting arterial blood pressure from non-invasive 4D flow MRI data using physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, 358, p.112623.(***Featured in the "Most Cited Articles" section of CMAME as one of the most cited papers of the years 2019-2023.***)

* *indicates equal contribution*

Workshops

AI for Science Track for AI+X Summit: Georgios Kissas with Vera Balmer and Sid Mishra, 2024. (https://www.plusx.ai)

2022 TRIPODS Winter School and Workshop on Interplay between Artificial Intelligence and Dynamical Systems: Paris Perdikaris with Georgios Kissas and Jacob Seidman, Supervised learning in function spaces, Mathematical Institute for Data Science, Johns Hopkins University, 2022. (https://github.com/PredictiveIntelligenceLab/TRIPODS_Winter_School_2022)

Minisymposia

"Biophysics-Informed Machine Learning", Organizer/Chair: Georgios Kissas, Co-chair: Jacob Seidman, 2023 Platform for Advanced Scientific Computing (PASC) Conference, Davos, Switzerland, 2023.

Academic Experience

ETH, Zurich, Switzerland
Lecturer

- Projects in Machine Learning Research, Spring 2025.
- Projects in Machine Learning Research, Spring 2024.

University of Pennsylvania, Philadelphia, Pennsylvania USA
Teaching Assistant

- ENM 360 Introduction to Data-Driven Modelling, Fall 2019-2020.
  Responsible for creating homework assignments, creating exams, teaching, and grading.
- MEAM 536 Viscous Fluid Flows Spring 2021.
  Responsible for recitations, and grading.

Project Supervision

- **Diego Renner**: "On differentiable simulations of haemodynamic systems", 2023, Co-supervised with Sid Mishra and Ben Moseley, ETHZ, Grade: 6.0, Published.
- **Levi Lingsch**: "Toward Personalized Cardiovascular Healthcare: Flow Matching Neural Operators for Forward and Inverse Problems", 2024, Co-supervised with Sid Mishra, ETHZ, Grade: 6.0, Published.
- **Orestis Oikonomou**: "Formal Grammar Foundation Models for PDE Discovery and Solution", Co-supervised with Sid Mishra and Davide Scaramuzza, UZH, 2024, Grade:6.0, Published.
- **Karin Yu**: "Formal Language for ODE Discovery", Co-supervised with Eleni Chatzi, ETHZ, Published.

Academic Service

Reviewer for: NeurIPS, ICML, ICLR, Journal of Machine Learning Research, Computer Methods in Applied Mechanics and Engineering, Journal of Scientific Computing, Nature Communications

ML Frameworks

- Jax, Pytorch